

L10: Some Empirical Findings About Cascades

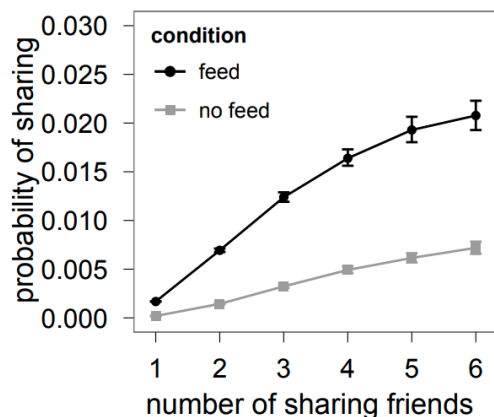


Image Source: ["The role of social networks in information diffusion"](https://dl.acm.org/doi/pdf/10.1145/2187836.2187907) by Bakshy et al

We summarize here some general findings that have been observed repeatedly in the last few years in the context of different social networks and experiments of social influence

The first such finding is that the probability of adoption seems to follow, at least in most cases, the diminishing returns model.

For instance, we include here a result from a large-scale (253 million Facebook users) randomized study by Eytan Bakshy and his collaborators. The main question was whether a user will share a Web link with his/her Facebook friends, depending on whether some of his/her friends also shared

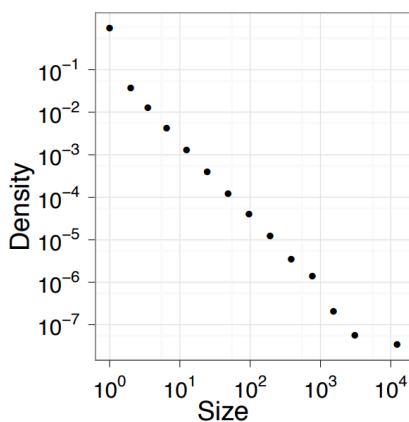
that link. The study also examined how this **"probability of link sharing"** depends on whether users were exposed to their friends' link-sharing behavior (**"feed"**) versus those that were not exposed to such information (**"no feed"**).

Note that in both cases the probability of adoption increases in a concave manner, providing support to the diminishing returns model.

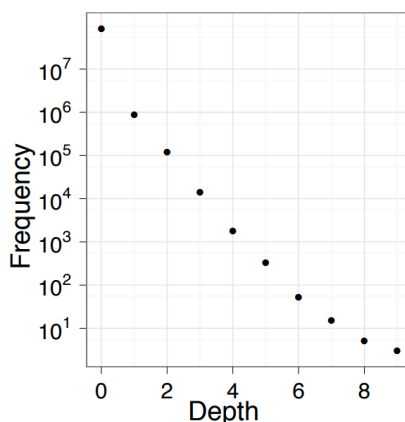
The authors also showed that those who are exposed to their friends' link-sharing behavior are significantly more likely to spread information and do so sooner than those who are not exposed.

The study also examined the relative role of strong and weak ties in information propagation. Although stronger ties are individually more influential, it is the more abundant weak ties that are responsible for the propagation of novel information.

Image Source: ["Everyone's an influencer: quantifying influence on Twitter"](https://dl.acm.org/doi/pdf/10.1145/1935826.1935845) by E. Bakshy et al.



(a) Cascade Sizes



(b) Cascade Depths

Another empirical finding that has been repeatedly shown by many studies of online social networks relates to the size and depth of cascades.

For instance, an observational study of Twitter data analyzed 74 million tweet cascades that mention a URL. The tweets originated from 1.6 million **"seed"** users during a two-month period in 2009.

A first major observation is that the vast majority (90%) of tweets do NOT trigger a network cascade. They stop at the source. An additional 9% propagate only to one

other user.

However, even though very few tweets trigger a significant network cascade, it is interesting that there are also tweets that cause major cascades, with a size that exceeds 1000 users and a depth of 8 or higher.

The distribution at the left shows that the cascade size follows a power-law distribution -- note that both axes are logarithmic and the function decreases almost linearly.

The distribution at the right shows that the cascade depth follows an exponential decrease -- note that the x-axis is linear while the y-axis is logarithmic and the decrease is almost linear.

These findings are not applicable only to Twitter cascades – similar results have been shown for most other diffusion phenomena in both offline and online social networks.

Food For Thought

1. Read the original paper mentioned in this page to see how the authors quantified the effect of strong versus weak ties.
2. The literature includes many more similar empirical results about the distribution of cascade size or cascade depth. Review the literature to find at least one more such reference.